

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

Group ID: 25-26J-109

Project Title: Multimodal AI Framework for Early Parkinson's Disease Detection

1. IT22093332: Wearable-based motion biomarker analysis for early detection of Parkinson's Disease
2. IT22551498: Cognitive Biomarker Analysis for Early Detection of Parkinson's Disease
3. IT22079336: Speech and vocal biomarker analysis for early detection of Parkinson's Disease
4. IT22884060: Drawing-based Parkinson's Disease risk prediction using spiral and wave drawing analysis

		IT22093332	IT22551498	IT22079336	IT22884060
Data Pipeline	Data Sources:	• PhysioNet PADS (Parkinson's Disease Smartwatch) dataset - 355 participants (276 PD, 79 healthy controls) recorded using Apple Watch Series 4 at 100 Hz. • Additional real-world validation data collected via custom ESP32/MPU-6050 wearable under informed consent.	• Primary dataset sourced from the Parkinson's Progression Markers Initiative (PPMI) Curated Data Cut, comprising 4,267 unique baseline patients across three cohorts: 336 Healthy Controls (COHORT=2) and 3,931 PD Spectrum patients — confirmed PD (COHORT=1) and prodromal (COHORT=4) • SWEDD participants (COHORT=3) excluded due to diagnostic ambiguity (DAT-negative parkinsonism with uncertain PD classification) • Analysis restricted to single baseline visit per participant (EVENT_ID="BL") to replicate	• Parkinson's Disease voice datasets obtained from the MDVR-KCL dataset and the UCI Machine Learning Repository. • The MDVR-KCL dataset contains reading and spontaneous speech recordings collected from individuals with Parkinson's Disease and healthy control participants. • The UCI dataset contains biomedical acoustic voice measurements extracted from sustained phonation recordings.	• Dataset obtained from a previously published study (Zhu et al., 2025) using smartphone-based touchscreen drawing data Finger drawing on smartphone screens enables early Parkinson's disease detection through hybrid 1D-CNN and BiGRU deep learning architecture

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			single-session screening deployment scenario and eliminate longitudinal data leakage - Target labels assigned as: Healthy Controls = 0, PD Spectrum (COHORT 1 + COHORT 4) = 1 - Dataset accessed via registered PPMI data portal under data use agreement - Additional real-world validation data collected from clinic participants under informed consent procedures for preliminary feasibility validation of the digitised cognitive task pipeline	Additional real-world validation data was collected from participants under informed consent procedures for testing and evaluation purposes.	PLOS One Subject demographic information Dataset - Additional real-world validation data was collected from participants under informed consent procedures for testing and evaluation purposes. - PaHaW, NewHandPD
	Data Preprocessing:	- Raw accelerometer and gyroscope signals windowed into non-overlapping 256-sample windows (2.56s at 100 Hz). - Per-axis mean subtraction applied to remove gravity components. - Eight statistical and spectral features extracted per sensor (RMS, Std Dev, Max, Min, Dominant Frequency, Tremor)	Cognitive Task Score Preprocessing: - SDMT scores linearly scaled from 60-second app window to 90-second PPMI standard: $\text{sdmttotal} = \text{correct_matches} \times 1.5$, maintaining commensurability with the training cohort. - TMT-A recorded as total completion time in seconds. Null assigned if trial	Acoustic Data Preprocessing: - Removed duplicate records and verified that no missing values were present. - Outlier handling performed using the Interquartile Range (IQR) capping method. - Stratified train-test split (80/20) applied to	- Coordinate sequences smoothed to reduce sensor noise and normalized using min-max scaling - Velocity, acceleration, and jerk computed via numerical differentiation

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		Band Power 3–7 Hz, Tremor Ratio, Spectral Entropy). - Three demographic features (age, sex, handedness) and three one-hot task-category flags appended. - All 17 numeric features standard-scaled using pre-computed constants embedded in model header.	incomplete (≥ 3-minute maximum), handled downstream by median imputation. - DVS_LNS (Letter-Number Sequencing) not implemented as an in-app task — passed as null at inference time and imputed from training-fold median. Feature Engineering (leak-free): - Age-adjusted TMT score: $tmt_age_adjusted = -TMT_A / (age / 60)$. Normalises psychomotor speed for age-related slowing, separating PD-specific slowing from normative decline. Computed using training-fold statistics only. - Cognitive dispersion index: $\sigma(z_SDMT, z_LNS, z_TMT-A)$. Standard deviation of z-scored cognitive task performance. Captures domain-specific asymmetry characteristic of early PD. - All z-scoring parameters (mean, SD) computed on training fold within each CV	preserve class distribution. - StandardScaler used for feature normalization, fitted only on training data. - Highly correlated features (Pearson correlation > 0.95) were removed to reduce redundancy. Linguistic Data Preprocessing: - Audio recordings were transcribed using OpenAI Whisper ASR model. - Outlier detection and capping performed using the IQR method. - Extreme values in linguistic features were capped at the 95th percentile where necessary. - Stratified splitting applied while maintaining both diagnosis class and speech type distribution. - StandardScaler applied to linguistic features for normalization.	- Irregular timestamps corrected using linear interpolation - Extracted features standardized to zero mean and unit variance using StandardScaler - Scaler fitted only on training data within each LOOCV fold to prevent data leakage
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			iteration and applied to validation/test folds without refitting. Imputation and Class Imbalance: - Missing values imputed via SimpleImputer (median strategy). Imputer fitted exclusively on training fold and applied to validation/test folds without refitting — strict leakage prevention. - Class imbalance (11.7:1) addressed via sample-weight compensation ($w_{\text{healthy}} = 11.7$, $w_{\text{spectrum}} = 1.0$). Synthetic oversampling (SMOTE) excluded to preserve clinical feature distributions and avoid physiologically implausible interpolated samples.		
	Data storage and versioning	Raw IMU windows processed on-device; result payloads (DONE:\<PD\ HP\>:\<ratio\>) streamed via BLE GATT to mobile client and stored per session in Google Firestore.	Structured Feature Storage: 9-feature vectors (SDMTOTAL, DVS_LNS, TMT_A, age_at_visit, SEX, fampd, rem, tmt_age_adjusted, cognitive_dispersion) stored in structured tabular format	- Processed datasets organized in structured tabular format. - Data securely stored using controlled-access cloud storage and local development environments.	Extracted 16-feature vectors organized into structured tabular format (CSV/dataframe) for model training and evaluation

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		Model artifacts (quantized .tflite, scaler constants) versioned via GitHub.	(Pandas DataFrame / CSV) for model training and evaluation. Production Data Storage: - Per-session cognitive results (risk_probability, percentile_rank, risk_level, contributing_factors) stored in Google Firestore (NoSQL), linked to authenticated user UID, enabling longitudinal result tracking. - Raw task metrics (TMT-A completion time, SDMT correct count, error counts) logged per session for audit and future retraining. Versioning and Reproducibility: - Model artefacts packaged as a single atomic joblib deployment bundle (calibrated model, imputer, engineering statistics, threshold, percentile thresholds) and versioned via GitHub. - Preprocessing statistics stored within the bundle to guarantee inference-time consistency with training-	- GitHub is used for version control and collaboration. - Preprocessed datasets and trained model artifacts versioned for reproducibility.	- Raw drawing sessions (x, y, timestamp sequences) stored in Google Firestore (NoSQL) for longitudinal tracking - Model prediction outputs (risk_percentage, risk_level, label, confidence) stored in Google Firestore per user session, enabling historical result retrieval and progression monitoring - Code and model versions managed via GitHub
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			time preprocessing prevents version mismatch between model weights and feature pipeline. Docker image versioned in GCP Artifact Registry. GitHub Actions CI/CD pipeline triggers rebuild and redeployment on every push to main.		
Model Development	Model Selection	- Quantized 1D Convolutional Neural Network (1D CNN) selected for on-device deployment. - Evaluated against hand-crafted feature pipelines; deep learning on raw IMU signals outperformed classical approaches. - Quantized to 8-bit integer precision using TensorFlow Lite for Microcontrollers — final model size 3.7 KB, inference latency 2.1 ms per window on ESP32.	Architecture Benchmarked (5-fold StratifiedGroupKFold CV): - Logistic Regression (L2-regularised, solver=lbfgs): CV AUC 0.7831 - Random Forest (100 estimators): CV AUC 0.8021 - XGBoost (scale_pos_weight=11.7): CV AUC 0.8268 - Gradient Boosting Machine (GBM): CV AUC 0.8310 - Stacking Ensemble (XGBoost + RF + LR meta-learner): CV AUC 0.8241 Final Selection:	Acoustic Models - Support Vector Machine (SVM) - Random Forest - XGBoost - Multi-layer Perceptron (MLP) Linguistic Models - Logistic Regression - Support Vector Machine (SVM) - Random Forest - XGBoost Best Performing Models Acoustic Model - Best model: SVM with RBF kernel.	Model Selection Phase 1: Deep learning models (CNN–BiGRU, ResNet50 fusion, QuadStream) were tested but abandoned due to overfitting on the small dataset (N = 56). Phase 2 (Final): Switched to feature-engineered models with better generalization: - Logistic Regression - SVM (RBF)

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			- XGBoost selected based on superior holdout generalisation (AUC 0.8011 vs GBM 0.7897), despite GBM marginally leading in CV. XGBoost's smaller CV→holdout gap (0.0257 vs GBM's 0.0413) confirms stronger generalisation to unseen data. - Deep learning architectures excluded: insufficient training sample for reliable convergence and incompatibility with per-user SHAP explainability requirement.	- Achieved high ROC-AUC performance on selected acoustic feature sets. Linguistic Model - Best model: Logistic Regression for classification accuracy. - SVM demonstrated strong ROC-AUC performance for linguistic classification tasks.	- Random Forest - Small MLP - Compact QuadStream variants (NB07, NB09, NB12, NB13)
	Model Training	5-fold Group K-Fold cross-validation with patient-level grouping to prevent window-level data leakage. 80/20 stratified train/test split at patient level. T - wo-tier thresholding: window-level threshold ($t_w = 0.560$) derived via Youden's J statistic; session-level threshold ($t_s =$	Cross-Validation Protocol: 5-fold StratifiedGroupKFold grouped by patient identifier (PATNO). Stratification preserves 11.7:1 class ratio; patient-level grouping ensures no patient appears in both training	Cross-validation using GridSearchCV and Stratified K-Fold validation was applied for hyperparameter optimization and model generalization. Model performance was evaluated using accuracy, precision, recall, F1-score,	- Leave-One-Out Cross-Validation (LOOCV) across all 58 subjects — each subject served as an independent test case - Evaluated three feature

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		0.44) via exhaustive grid search maximizing balanced accuracy. Temporal aggregation over 12 consecutive windows (~30 seconds) per session.	and validation folds simultaneously. - All preprocessing (imputation, z-scoring, feature engineering) fitted on training fold only and applied to validation fold without refitting enforcing strict end-to-end leakage prevention. Hyperparameter Tuning: - RandomizedSearchCV: XGBoost tuned over 150 iterations, GBM over 100 iterations. 3-fold inner CV. Search space: <code>n_estimators</code> (100–1000), <code>max_depth</code> (3–8), <code>learning_rate</code> (0.01–0.30), <code>subsample</code>, <code>colsample_bytree</code>, <code>min_child_weight</code>, <code>reg_alpha</code>, <code>reg_lambda</code>, <code>gamma</code>. Calibration and Threshold Selection: - Platt scaling applied post-training (CalibratedClassifierCV,	ROC-AUC, and confusion matrices. - For the acoustic modality, the SVM model with an RBF kernel achieved the best performance with strong ROC-AUC results. For the linguistic modality, Logistic Regression achieved the highest classification accuracy, while SVM showed competitive ROC-AUC performance. - SHAP analysis was used to identify important biomarkers influencing predictions. Key acoustic predictors included <code>spread1</code>, <code>voice_health_score</code>, <code>NHR</code>, and <code>instability_ratio</code>, while important linguistic predictors included <code>noun_ratio</code>, <code>repetition_ratio</code>, and <code>parse_tree_depth_avg</code>.	configurations: geometric-only, kinematic-only, combined 16-feature (combined consistently superior) - Hyperparameter tuning via systematic search (model size, dropout, regularization) - Model selection based on weighted composite score: Recall $\times 0.35$, AUC $\times 0.30$, Accuracy $\times 0.20$, Specificity $\times 0.15$ (Recall prioritized to minimize false negatives in clinical screening) - Monte Carlo Dropout used at inference time for
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			sigmoid, cv=5). Brier Score improved from 0.180 (uncalibrated) to 0.140 (calibrated) essential for valid multimodal late fusion. - Optimal threshold 0.4364 derived via Youden's J on out-of-fold CV predictions (not holdout), preventing threshold overfitting. Holdout Results: - AUC-ROC: 0.8011 Sensitivity: 0.7649 Specificity: 0.6119 Balanced Accuracy: 0.6884 - SHAP top features: fampd (1.15), rem (0.52), tmt_age_adjusted (0.36), age (0.21), SDMTOTAL (0.18)		uncertainty estimation
Model Integration	Tools used	- VSCode for backend and model development - TensorFlow Lite for Microcontrollers, ESP32 (Arduino/C++ firmware), MPU-6050 IMU, BLE GATT protocol	- Python 3.11 — scikit-learn, XGBoost, SHAP, pandas, numpy, joblib - Google Colab — model training, hyperparameter search, SHAP analysis	- VSCode for backend and model development - Google Colab for model training, hyperparameter search, SHAP analysis	- VSCode for backend and model development - Google Colab for model training,

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		• Google Colab for model training, hyperparameter search, SHAP analysis • Python for machine learning implementation • FastAPI for backend services • React Native for mobile frontend development • Firebase for database management • Firebase Authentication for JWT token-based user identity management • Google Cloud Run for serverless model deployment (auto-scaling, HTTPS) • GCP Artifact Registry for versioned Docker image storage • Docker (Python 3.10-slim) - backend containerization • GitHub for version control and collaboration	• VSCode — backend and mobile development • FastAPI — REST inference API endpoint • React Native (Expo) — mobile frontend, TMT-A and SDMT task implementation • Firebase Firestore — user profile and session result storage • Firebase Authentication — JWT token-based user identity management • Google Cloud Run — serverless model deployment (auto-scaling, HTTPS) • GCP Artifact Registry — versioned Docker image storage • Docker (Python 3.10-slim) — backend containerization • GitHub — version control, CI/CD via GitHub Actions	• Python for machine learning implementation • FastAPI for backend services • React Native for mobile frontend development • Firebase for database management • Firebase Authentication for JWT token-based user identity management • Google Cloud Run for serverless model deployment (auto-scaling, HTTPS) • GCP Artifact Registry for versioned Docker image storage • Docker (Python 3.10-slim) - backend containerization • GitHub for version control and collaboration	• hyperparameter search, SHAP analysis • Python for machine learning implementation • FastAPI for backend services • React Native for mobile frontend development • Firebase for database management • Firebase Authentication for JWT token-based user identity management • Google Cloud Run for serverless model deployment (auto-scaling, HTTPS) • GCP Artifact Registry for versioned Docker image storage
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					• Docker (Python 3.10-slim) - backend containerization • GitHub for version control and collaboration
Model Deployment	Testing Environments	• Validated on PADS held-out test partition. Patient-level balanced accuracy: 70.2% (95% CI: 63.8%–75.4%); AUC: 0.732 (95% CI: 0.694–0.816); Sensitivity: 74.3%; Specificity: 65.8%. • Bootstrap resampling (n=10,000 iterations) applied for confidence interval estimation.	Dataset Validation: • Primary evaluation on stratified 20% held-out PPMI test set (n=854: 787 PD Spectrum, 67 Healthy Controls). Held-out set sequestered throughout all training, tuning, and threshold selection; accessed exactly once for final performance reporting. • Preliminary clinical validation at a neurology clinic: confirmed PD patients and healthy controls completed digitised TMT-A and SDMT tasks on a smartphone device under researcher supervision. Software Testing: • Unit tests (pytest): feature engineering transformers, imputer leakage prevention, SDMT 1.5× scaling formula,	• The system was tested using both publicly available datasets and real-world participant data collected during the validation phase. • Validation included data collected from individuals with Parkinson’s Disease symptoms and healthy participants under informed consent procedures. • Acoustic and linguistic predictions were evaluated using real patient speech recordings and spontaneous speech samples. • Model outputs were analyzed using classification metrics such as accuracy, precision,	• Evaluated using a smartphone touchscreen dataset (58 participants: 28 PD, 30 healthy controls) and further validated with real-world patient drawing tests, where individuals performed spiral and wave tasks on the device to assess practical performance. • Validation: Leave-One-Out Cross-Validation (LOOCV) to ensure subject-level

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			TMT-A null assignment, API input validation (Pydantic schema). - Integration tests: full inference pipeline from JSON request to JSON response, Firestore demographic retrieval (mocked), Firebase JWT authentication enforcement (HTTP 401 on failure). - Load testing (Locust): p95 API response latency <500ms sustained under 50 concurrent users on Cloud Run. Evaluation Metrics: - AUC-ROC, Sensitivity, Specificity, Balanced Accuracy, Brier Score, calibration reliability diagram, confusion matrix at threshold=0.4364.	recall, F1-score, ROC-AUC, and confusion matrices. Feedback and observations from the validation process were used to assess the practical applicability and robustness of the multimodal screening framework.	independence and prevent data leakage. - Metrics: Accuracy, Recall, Specificity, ROC-AUC, and confusion matrix, with a weighted score prioritizing Recall (to reduce false negatives). - Testing: Unit, integration, and system testing (25 cases, 80% pass rate), including edge cases such as empty input and device-related issues. - User Evaluation: User Acceptance Testing (5 participants) showed strong usability, with an average score of 4.5/5.0.
	Deployment Platform	On-device inference running on ESP32 microcontroller	Backend:	Backend:	Backend:

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		(TinyML). BLE GATT communication to React Native mobile client Backend: - Google Cloud Run (serverless, auto-scaling 1–10 instances). HTTPS enforced at ingress. Firebase Auth JWT verification on all endpoint calls. Minimum 1 warm instance maintained to eliminate cold-start latency under real traffic. Memory: 2GB per instance; CPU: 2 vCPUs. Frontend: - React Native (Expo) targeting iOS and Android. Data Services: - Firebase Authentication for JWT-based user identity management. - Google Cloud Firestore for user profile data and per-session wearable result persistence	- Google Cloud Run (serverless, auto-scaling 1–10 instances). HTTPS enforced at ingress. Firebase Auth JWT verification on all endpoint calls. Minimum 1 warm instance maintained to eliminate cold-start latency under real traffic. Memory: 2GB per instance; CPU: 2 vCPUs. Frontend: - React Native (Expo) targeting iOS and Android. TMT-A and SDMT rendered as native touchscreen interactions with millisecond-precision event logging. Data Services: - Firebase Authentication for JWT-based user identity management. - Google Cloud Firestore for user profile data and per-session cognitive result persistence.	- Google Cloud Run (serverless, auto-scaling 1–10 instances). HTTPS enforced at ingress. Firebase Auth JWT verification on all endpoint calls. Minimum 1 warm instance maintained to eliminate cold-start latency under real traffic. Memory: 2GB per instance; CPU: 2 vCPUs. Frontend: - React Native (Expo) targeting iOS and Android. Speech tasks recorded through smartphone microphones with real-time audio capture and timestamp logging for acoustic and linguistic biomarker extraction. Data Services: - Firebase Authentication for JWT-based user identity management. - Google Cloud Firestore for user profile data and	- Google Cloud Run (serverless, auto-scaling 1–10 instances). HTTPS enforced at ingress. Firebase Auth JWT verification on all endpoint calls. Minimum 1 warm instance maintained to eliminate cold-start latency under real traffic. Memory: 2GB per instance; CPU: 2 vCPUs. Frontend: - React Native (Expo) targeting iOS and Android. TMT-A and SDMT rendered as native touchscreen interactions with millisecond-precision event logging for
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				per-session voice result persistence	handwriting and drawing kinematic analysis. Data Services: • Firebase Authentication for JWT-based user identity management. • Google Cloud Firestore for user profile data and per-session drawing result persistence
	Deployment Method	• 1D CNN quantized to 8-bit via TensorFlow Lite for Microcontrollers and flashed directly onto ESP32 firmware. • Scaler constants embedded in model header. No cloud dependency for inference - fully offline capable. • Backend containerized using Docker (Python 3.10-slim) with model weights and feature scalers included.	Containerization: • Backend containerized using Docker (Python 3.10-slim). Atomic joblib deployment bundle included in image at build time: calibrated XGBoost model, Platt scaling wrapper, SimpleImputer, engineering statistics, classification threshold, percentile thresholds.	• Backend containerized using Docker (Python 3.10-slim) with model weights and feature scalers included. • Models loaded at startup to avoid cold-start issues under real traffic. • GitHub Actions pipeline: on push to main branch, Docker image is rebuilt, pushed to GCP Artifact	• Backend containerized using Docker (Python 3.10-slim) with model weights and feature scalers included. • Models loaded at startup to avoid cold-start issues under real traffic.

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		GitHub Actions pipeline: on push to main branch, Docker image is rebuilt, pushed to GCP Artifact Registry, and deployed to Cloud Run with zero-downtime rolling update	- Atomic bundling prevents model-preprocessing version mismatch a common silent failure mode in deployed ML systems. CI/CD: - GitHub Actions pipeline: on push to main branch, Docker image is rebuilt, pushed to GCP Artifact Registry, and deployed to Cloud Run with zero-downtime rolling update. Inference Endpoint: - POST /api/cognitive-analysis/predict - Input: {sdmtotal, tmt_a, dvs_ins, age_at_visit, SEX, fampd, rem} - Output: {risk_probability, percentile_rank, risk_level, contributing_factors[]} - Median p95 latency: <200ms including Firestore demographic retrieval.	Registry, and deployed to Cloud Run with zero-downtime rolling update	GitHub Actions pipeline: on push to main branch, Docker image is rebuilt, pushed to GCP Artifact Registry, and deployed to Cloud Run with zero-downtime rolling update
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Future Enhancements	Model Improvement:	• Cross-device IMU calibration across different wearable hardware. • PPG-based heart rate variability integration for autonomic dysfunction assessment. • Federated learning for model updates without compromising patient data privacy. • Prospective validation with demographically diverse cohorts.	• Incorporate richer digital biomarkers from the TMT-A mobile task like inter-tap intervals, path deviation, hesitation patterns, and lift-off timestamps to supplement the current score-based features and capture fine-grained motor-cognitive signatures • Expand training data with larger, demographically diverse cohorts beyond PPMI to improve generalization across different age groups, ethnicities, and disease severity levels • Investigate temporal cognitive monitoring by leveraging longitudinal Firestore session history to detect subtle cognitive changes across successive assessments over time • Explore multimodal feature-level fusion combining cognitive embeddings directly with voice, drawing, and wearable features rather than late fusion of	• Exploring advanced multimodal fusion techniques such as feature-level fusion, ensemble-based decision fusion, and deep learning-based architectures. • Increasing dataset diversity across different demographics, languages, and disease severity levels. • Collecting larger real-world speech datasets to improve model generalization and robustness. • Engineering additional acoustic and linguistic biomarkers with stronger discriminative capability. • Investigating temporal speech dynamics and conversational behavior features for improved detection performance. • Implementing automated model monitoring and performance tracking for	• Cross-device validation: prospective data collection across multiple smartphone models and demographic groups to formally validate generalization • Longitudinal progression tracking: leverage stored Firestore session history to detect subtle motor changes over time • Multimodal risk fusion: integrate wearable sensor, voice, and cognitive assessment streams into a unified risk score • Expand dataset to larger, geographically
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			independent probability scores Integrate prediction drift monitoring (e.g. Evidently AI) to detect distributional shifts in incoming cognitive task scores, triggering automated retraining alerts when model performance degrades below a defined AUC threshold	continuous evaluation after deployment.	diverse cohorts to improve statistical power
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